

ZHAORUN CHEN

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EDUCATION

University of Chicago Ph.D. in Computer Science – Advisor: Prof. Bo Li and Prof. Dawn Song	2024.09 - 2026.12
Purdue University M.S in Computer Engineering	2022.08 - 2023.08
Shanghai Jiao Tong University B.E in Computer Engineering	2018.09 - 2022.06

WORK EXPERIENCES

Meta Senior Research Scientist (IC5), Meta Superintelligence Labs	2026.07–Present
- Tech lead for red teaming on Bo Li and Dawn Song’s team, leading AutoRT (i.e., automated red-teaming) efforts for Muse Spark 1.1–1.3, Muse Agent, and Muse Code to provide in-depth risk assessments prior to release. - Led the development of the first systematic AutoRT platform at Meta MSL, including 50+ realistic simulation environments (Muse connectors and browser-based environments) and a fully end-to-end automated red-teaming agent. Building on my first-authored work DTap, the platform now serves as Meta’s central infrastructure for ongoing red teaming across multiple Muse products. - Led indirect prompt injection (IPI) and cybersecurity red-teaming evals for Muse safety evaluation reports, in collaboration with Dan Hendrycks and Sail Wang. - Led the development of training presets to mitigate destructive behavior in benign coding tasks in Muse Spark 1.3, in collaboration with Andy Zou, Vincent Zhao, and Alex Pan; the resulting model outperforms other SOTA models on SABER by over 10%. - Developed the browser action guard for Muse to detect high-risk submission and payment actions in browser CVMs under prompt injection, achieving over 80% recall with less than 0.1% FPR across a continuous 72-hour window of Muse user traffic.	
Virtue AI Tech Lead of Agent Red-Teaming, Founding Member of Technical Staff	2024.01 - 2026.06
- Led the development of Virtue AgentSuite, Virtue Agent ForgingGround, Virtue Action Guard, Virtue AgentGateway	
Meta Research Scientist Intern working with Jason Weston at Meta Superintelligence Labs	2025.06 - 2025.10
- Led the research of the first synthetic training framework for general AI agents (DreamGym, Early Experience)	

SELECTED PUBLICATIONS

Publication summary: **40+ papers in top AI conferences across NeurIPS (12), ICLR (11), and ICML (4); 2,600+ total citations; 1,500+ first-author citations; 700+ citations for the most-cited paper.** Full publication list available on [Google Scholar](#).

- [1] **Zhaorun Chen** and the Meta MSL Preparedness, Red Teaming, and Alignment Team, AI Security Team, [Muse Spark 1.3 & Muse Agent Evaluation Report](#). [Paper] [\[Red-Teaming\]](#)
- [2] **Zhaorun Chen** and the Meta MSL Preparedness, Red Teaming, and Alignment Team, AI Security Team, [Muse Spark 1.2 & Muse Code Evaluation Report](#). [Paper] [\[Red-Teaming\]](#)
- [3] **Zhaorun Chen** and the Meta MSL Preparedness, Red Teaming, and Alignment Team, AI Security Team, [Muse Spark 1.1 Evaluation Report](#). [Paper] [\[Red-Teaming\]](#)
- [4] **Zhaorun Chen**, Xun Liu, Haibo Tong, Chengquan Guo, Yuzhou Nie, Jiawei Zhang, Mintong Kang, Chejian Xu, Qichang Liu, Xiaogeng Liu, Tianneng Shi, Chaowei Xiao, Sanmi Koyejo, Percy Liang, Wenbo Guo, Dawn Song, and Bo Li, [DecodingTrust-Agent Platform \(DTap\): A Controllable and Interactive Red-Teaming Platform for AI Agents](#), Arxiv 2026. [Paper] [\[Agent Red-Teaming\]](#)
- [5] **Zhaorun Chen**, Zhen Xiang, Chaowei Xiao, Dawn Song, Bo Li, [AgentPoison: Red-teaming LLM Agents via Poisoning Memory or Knowledge Bases](#), in Proceeding of the Thirty-Eighth Conference on Neural Information Processing Systems (**NeurIPS 2024**), Vancouver, Canada, Dec 2024. [Paper] [\[Code\]](#) [\[LLM Agent Safety\]](#)
- [6] **Zhaorun Chen**, Mintong Kang, Bo Li, [ShieldAgent: Shielding Agents via Verifiable Safety Policy Reasoning](#), in Proceedings of the Forty-Second International Conference on Machine Learning (**ICML 2025**), Vancouver, Canada, July 2025. [Paper] [\[Code\]](#) [\[Agent Safety\]](#) [\[Guardrail\]](#)
- [7] **Zhaorun Chen**, Zhuokai Zhao, Hongyin Luo, Huaxiu Yao, Bo Li, Jiawei Zhou, [HALC: Object Hallucination Reduction via Adaptive Focal-Contrast Decoding](#), in Proceeding of the Forty-first International Conference on Machine Learning (**ICML 2024**), Vienna, Austria, July 2024. [Paper] [\[Code\]](#) [\[Multi-modal Hallucination\]](#)
- [8] **Zhaorun Chen**, Francesco Pinto, Minzhou Pan, Bo Li, [SafeWatch: An Efficient Safety-Policy Following Video Guardrail Model with Transparent Explanations](#), in Proceedings of the 13th International Conference on Learning Representations (**ICLR 2025**), Singapore, Apr 2025. [Paper] [\[Code\]](#) [\[Video Safety Reasoning\]](#) [\[RL Post-Training\]](#)
- [9] **Zhaorun Chen**, Xun Liu, Mintong Kang, Jiawei Zhang, Minzhou Pan, Shuang Yang, Bo Li, [ARMs: Adaptive Red-Teaming Agent against Multimodal Models with Plug-and-Play Attacks](#), in Proceedings of the 14th International Conference on Learning Representations (**ICLR 2026**), Rio de Janeiro, Brazil, April 2026. [Paper] [\[Red-Teaming\]](#)[\[Multi-Modal\]](#)
- [10] **Zhaorun Chen**, Zhuokai Zhao, Kai Zhang, Bo Liu, Qi Qi, Yifan Wu, Tarun Kalluri, Sara Cao, Yuanhao Xiong, Haibo Tong, Huaxiu Yao, Hengduo Li, Jiacheng Zhu, Xian Li, Dawn Song, Bo Li, Jason Weston, Dat Huynh, [Scaling Agent Learning via Experience Synthesis](#), in Proceedings of the 14th International Conference on Learning Representations (**ICLR 2026**), Rio de Janeiro, Brazil, April 2026. [Paper] [\[Agent RL\]](#)
- [11] **Zhaorun Chen**, Zichen Wen, Yichao Du, Yiyang Zhou, Chenhang Cui, Zhenzhen Weng, Haoqin Tu, Chaoqi Wang, Zhengwei Tong, Qinglan Huang, Canyu Chen, Qinghao Ye, Zhihong Zhu, Yuqing Zhang, Jiawei Zhou, Zhuokai Zhao, Rafael Rafailov, Chelsea Finn, Huaxiu Yao, [MJ-Bench: Is Your Multimodal Reward Model Really a Good Judge for Text-to-Image Generation?](#), in Proceeding of the Thirty-Ninth Conference on Neural Information Processing Systems (**NeurIPS 2025**), San Diego, the United States, Dec 2025. [Paper] [\[Code\]](#) [\[RL Post-Training\]](#)
- [12] **Zhaorun Chen**, Zhuokai Zhao, Zhihong Zhu, Ruiqi Zhang, Xiang Li, Bhiksha Raj, Huaxiu Yao, [AutoPRM: Automating Procedural Supervision for Multi-Step Reasoning via Controllable Question Decomposition](#), in Proceeding of 2024 Annual Conference of the North American Chapter of the Association for Computational Linguistics (**NAACL 2024**), Mexico City, Mexico, Jun 2024. [Paper] [\[Code\]](#) [\[RL Post-Training\]](#)
- [13] **Zhaorun Chen**, Zhuokai Zhao, Tairan He, Binhao Chen, Xuhao Zhao, Liang Gong, Chengliang Liu, [Safe Reinforcement Learning via Hierarchical Adaptive Chance-Constraint Safeguards](#), in Proceeding of 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS 2024**), Abu Dhabi ,UAE, October 2024. [Paper] [\[Code\]](#) [\[Safe RL\]](#) [\[Robotics\]](#)

- [14] Canyu Chen, Kangyu Zhu, **Zhaorun Chen**, Zhanhui Zhou, Shizhe Diao, Yiping Lu, Tian Li, Manling Li, and Dawn Song, [Is Decentralized LLM Agent RL Robust to Heterogeneity? An Asymmetric Tale](#), in Proceeding of The Fortieth Annual Conference on Neural Information Processing Systems (**NeurIPS 2026**), Sydney, Australia, Dec 2026. [Paper] [Agent RL]
- [15] Zhijian Zhou, Zesheng Ye, **Zhaorun Chen**, Bo Li, and Feng Liu, [CELEUS: Certifiable and Efficient LLM Evaluation via E-Processes](#), in Proceeding of The Fortieth Annual Conference on Neural Information Processing Systems (**NeurIPS 2026**), Sydney, Australia, Dec 2026. [Paper] [Agent Eval]
- [16] Yuzhou Nie, Ruilin Zhou, **Zhaorun Chen**, Jingyang Zhang, Yan Shao, Hongwei Li, Bo Li, Dawn Song, and Wenbo Guo, [NitroBox: Lightning-Fast Sandbox for Large-Scale RL Training](#), in Proceeding of The Fortieth Annual Conference on Neural Information Processing Systems (**NeurIPS 2026**), Sydney, Australia, Dec 2026. [Paper] [Agent RL]
- [17] Mintong Kang, **Zhaorun Chen**, Bo Li, [C-SafeGen: Certified Safe LLM Generation with Claim-Based Streaming Guardrails](#), in Proceeding of the Thirty-Ninth Conference on Neural Information Processing Systems (**NeurIPS 2025**), San Diego, the United States, Dec 2025. [Paper] [Code] [Guardrail]
- [18] Mintong Kang, **Zhaorun Chen**, Chejian Xu, Jiawei Zhang, Chengquan Guo, Minzhou Pan, Ivan Revilla, Yu Sun, Bo Li, [PolyGuard: Massive Multi-Domain Safety Policy-Grounded Guardrail Dataset](#), in Proceeding of the Thirty-Ninth Conference on Neural Information Processing Systems (**NeurIPS 2025**), San Diego, the United States, Dec 2025. [Paper] [Code] [Guardrail] [Dataset]
- [19] Chejian Xu, Jiawei Zhang, **Zhaorun Chen**, Chulin Xie, Mintong Kang, Zhuowen Yuan, Zidi Xiong, Chenhui Zhang, Lingzhi Yuan, Yi Zeng, Peiyang Xu, Chengquan Guo, Andy Zhou, Jeffrey Ziwei Tan, Zhun Wang, Alexander Xiong, Xuandong Zhao, Yu Gai, Francesco Pinto, Yujin Potter, Zhen Xiang, Zinan Lin, Dan Hendrycks, Dawn Song, Bo Li, [MMDT: Decoding the Trustworthiness and Safety of Multimodal Foundation Models](#), in Proceedings of the 13th International Conference on Learning Representations (**ICLR 2025**), Singapore, Apr 2025. [Paper] [Code] [Multi-modal Safety]
- [20] Chengquan Guo, Chulin Xie, Yu Yang, **Zhaorun Chen**, Zinan Lin, Xander Davies, Yarin Gal, Dawn Song, Bo Li, [RedCodeAgent: Automatic Red-teaming Agent against Diverse Code Agents](#), in Proceedings of the 14th International Conference on Learning Representations (**ICLR 2026**), Rio de Janeiro, Brazil, April 2026. [Paper] [Red-Teaming][Coding Agent]
- [21] Zichen Wen, Jiashu Qu, **Zhaorun Chen**, Dongrui Liu, Zhiyuan Liu, Ruixi Wu, Yicun Yang, Xiangqi Jin, Haoyun Xu, Xuyang Liu, Weijia Li, Chaochao Lu, Jing Shao, Conghui He, Linfeng Zhang, [The Devil behind the mask: An emergent safety vulnerability of Diffusion LLMs](#) (**ICLR 2026**), Rio de Janeiro, Brazil, April 2026. [Paper] [Red-Teaming][Diffusion LLMs]
- [22] Zixin Rao, Wentian Zhu, Chan Aristella Lu, **Zhaorun Chen**, Wei Niu, Le Guan, Bo Li, and Zhen Xiang, [FragFuse: Bypassing Access Control of Large Language Model Agents via Memory-Based Query Fragmentation and Fusion](#), in Proceeding of The 35th USENIX Security Symposium (**USENIX 2026**), Baltimore, MD, the United States, Aug 2026. [Paper] [Agent Red-Teaming]
- [23] Bo Liu, Simon Yu, Yiding Jiang, Ao Qu, Andrew Zhao, Zichen Liu, Junsu Kim, Zijian Zhou, Seungone Kim, Tongzheng Ren, Mickel Liu, Hanfei Yu, **Zhaorun Chen**, Weiyan Shi, Paul Pu Liang, Luke Zettlemoyer, Yejin Choi, Natasha Jaques, [SPADE: Self-play in adaptive synthetic executable environments](#), Arxiv 2026. [Paper] [Agent RL]
- [24] Zhanhao Hu, Dennis Jacob, Xiao Huang, **Zhaorun Chen**, Bo Li, and David Wagner, [Twin Agent: Context Residual Compression for Privilege Separated Agents](#), Arxiv 2026. [Paper] [Secured Agent by Design]
- [25] Kai Zhang, Xiangchao Chen, Bo Liu, Tianci Xue, Zeyi Liao, Zhihan Liu, Xiyao Wang, Yuting Ning, **Zhaorun Chen**, Xiaohan Fu, Jian Xie, Yuxuan Sun, Boyu Gou, Qi Qi, Zihang Meng, Jianwei Yang, Ning Zhang, Xian Li, Ashish Shah, Dat Huynh, Hengduo Li, Zi Yang, Sara Cao, Lawrence Jang, Shuyan Zhou, Jiacheng Zhu, Huan Sun, Jason Weston, Yu Su, Yifan Wu, [Agent Learning via Early Experience](#), in Proceedings of the Forty-Third International Conference on Machine Learning (**ICML 2026**), Seoul, South Korea, July 2026. [Paper] [Agent RL]

- [26] Nuoya Xiong, Yuhang Zhou, Hanqing Zeng, **Zhaorun Chen**, Furong Huang, Shuchao Bi, Lizhu Zhang, Zhuokai Zhao, [Token-Level LLM Collaboration via FusionRoute](#), in Proceedings of the Forty-Third International Conference on Machine Learning (**ICML 2026**), Seoul, South Korea, July 2026. [\[Paper\]](#) [\[LLM Infra\]](#)
- [27] Zijian Zhang, Kaiyuan Zheng, **Zhaorun Chen**, Joel Jang, Yi Li, Chaoqi Wang, Mingyu Ding, Dieter Fox, and Huaxiu Yao, [GRAPE: Generalizing Robot Policy via Preference Alignment](#), in Proceeding of the IEEE International Conference on Robotics and Automation (**ICRA 2026**), Vienna, Austria, June 2026. [\[Paper\]](#) [\[Code\]](#) [\[RL Post-Training\]](#) [\[Robotics\]](#)
- [28] Siwei Han, Haonian Ji, Siyang Xin, Juanquan Shi, Shi Qiu, Xinyu Ye, Peng Xia, Jiaqi Liu, **Zhaorun Chen**, Yiyang Zhou, Linjie Li, Lijuan Wang, Huaxiu Yao, [Paper2Figure: A Multi-Agent Collaborative System for Figure Generation Towards Academic Research Paper](#), in Proceeding of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2026**), Denver, CO, June 2026. [\[Paper\]](#) [\[Code\]](#) [\[Image Generation\]](#)
- [29] Haibo Tong, Zhaoyang Wang, **Zhaorun Chen**, Haonian Ji, Shi Qiu, Siwei Han, Zhongkai Xue, Yiyang Zhou, Peng Xia, Kexin Geng, Mingyu Ding, Rafael Rafailov, Chelsea Finn, Huaxiu Yao, [MJ-Video: Fine-Grained Benchmarking and Rewarding Video Preferences in Video Generation](#), in Proceeding of the Thirty-Eighth Conference on Neural Information Processing Systems (**NeurIPS 2025**), San Diego, the United States, Dec 2025. [\[Paper\]](#) [\[Code\]](#) [\[RL Post-Training\]](#)
- [30] Andy Zhou, Kevin Wu, Francesco Pinto, **Zhaorun Chen**, Yi Zeng, Yu Yang, Shuang Yang, Sanmi Koyejo, James Zou, Bo Li, [AutoRedTeamer: Autonomous Red Teaming with Lifelong Attack Integration](#), in Proceeding of the Thirty-Ninth Conference on Neural Information Processing Systems (**NeurIPS 2025**), San Diego, the United States, Dec 2025. [\[Paper\]](#) [\[Code\]](#) [\[Red-Teaming\]](#)
- [31] Shaobo Wang, Yantai Yang, Guo Chen, Peiru Li, Kaixin Li, Yufa Zhou, **Zhaorun Chen**, Linfeng Zhang, [Grounding and Enhancing Informativeness and Utility in Dataset Distillation \(ICLR 2026\)](#), Rio de Janeiro, Brazil, April 2026. [\[Paper\]](#) [\[Multi-Modal\]](#)[\[Data Distillation\]](#)
- [32] Haonian Ji, Shi Qiu, Siyang Xin, Siwei Han, **Zhaorun Chen**, Dake Zhang, Hongyi Wang, Huaxiu Yao, [From EduVisBench to EduVisAgent: A Benchmark and Multi-Agent Framework for Reasoning-Driven Pedagogical Visualization \(ICLR 2026\)](#), Rio de Janeiro, Brazil, April 2026. [\[Paper\]](#) [\[Multi-Modal\]](#)[\[Benchmark\]](#)
- [33] Yixiong Fang, Ziran Yang, **Zhaorun Chen**, Zhuokai Zhao, Jiawei Zhou, [Enhancing Vision-Language Model Reliability with Uncertainty-Guided Dropout Decoding](#), in Proceeding of the Thirty-Ninth Conference on Neural Information Processing Systems (**NeurIPS 2025**), San Diego, the United States, Dec 2025. [\[Paper\]](#) [\[Code\]](#) [\[Hallucination\]](#)
- [34] Zichen Wen, Shaobo Wang, Yufa Zhou, Junyuan Zhang, Qintong Zhang, Yifeng Gao, **Zhaorun Chen**, Bin Wang, Weijia Li, Conghui He, Linfeng Zhang, [Efficient Multi-modal Large Language Models via Progressive Consistency Distillation](#), in Proceeding of the Thirty-Ninth Conference on Neural Information Processing Systems (**NeurIPS 2025**), San Diego, the United States, Dec 2025. [\[Paper\]](#) [\[Code\]](#) [\[Multimodal Understanding\]](#)
- [35] Yiming Zhang, Zhuokai Zhao, **Zhaorun Chen**, Zhili Feng, Zenghui Ding, Yining Sun, [RankClip: Ranking-Consistent Language-Image Pretraining](#), in Proceeding of the International Conference on Computer Vision (**ICCV 2025**), Honolulu, the United States, Dec 2025. [\[Paper\]](#) [\[Code\]](#) [\[Multimodal Pre-training\]](#)
- [36] Yiming Zhang, Zhuokai Zhao, **Zhaorun Chen**, Zenghui Ding, Xianjun Yang, Yining Sun, [Beyond Training: Dynamic Token Merging for Zero-Shot Video Understanding](#), in Proceeding of the International Conference on Computer Vision (**ICCV 2025**), Honolulu, the United States, Dec 2025. [\[Paper\]](#) [\[Code\]](#) [\[Multimodal Understanding\]](#)
- [37] Peng Xia, Siwei Han, Shi Qiu, Yiyang Zhou, Zhaoyang Wang, Wenhao Zheng, **Zhaorun Chen**, Chenhang Cui, Mingyu Ding, Linjie Li, Lijuan Wang, Huaxiu Yao, [MMIE: Massive Multimodal Interleaved Comprehension Benchmark for Large Vision-Language Models](#), in Proceedings of the 13th Interna-

- tional Conference on Learning Representations (**ICLR 2025**) (**Oral Presentation**), Singapore, Apr 2025. [\[Paper\]](#) [\[Code\]](#) [\[Multi-modal Reasoning\]](#)
- [38] Yiyang Zhou, Zhaoyang Wang, Tianle Wang, Shangyu Xing, Peng Xia, Bo Li, Kaiyuan Zheng, Zijian Zhang, **Zhaorun Chen**, Wenhao Zheng, Xuchao Zhang, Chetan Bansal, Weitong Zhang, Ying Wei, Mohit Bansal, Huaxiu Yao, [AnyPrefer: An Automatic Framework for Preference Data Synthesis](#), in Proceedings of the 13th International Conference on Learning Representations (**ICLR 2025**), Singapore, Apr 2025. [\[Paper\]](#) [\[RL Post-Training\]](#)
- [39] Chenhang Cui, An Zhang, Yiyang Zhou, **Zhaorun Chen**, Gelei Deng, Huaxiu Yao, Tat-Seng Chua, [Fine-Grained Verifiers: Preference Modeling as Next-token in Vision-Language Alignment](#), in Proceedings of the 13th International Conference on Learning Representations (**ICLR 2025**), Singapore, Apr 2025. [\[Paper\]](#) [\[RL Post-Training\]](#)
- [40] Yiyang Zhou, Zhiyuan Fan, Dongjie Cheng, Sihan Yang, **Zhaorun Chen**, Chenhang Cui, Xiyao Wang, Yun Li, Linjun Zhang, Huaxiu Yao, [Calibrated Self-Rewarding Vision Language Models](#), in Proceeding of the Thirty-Eighth Conference on Neural Information Processing Systems (**NeurIPS 2024**), Vancouver, Canada, Dec 2024. [\[Paper\]](#) [\[Code\]](#) [\[RL Post-Training\]](#)
- [41] Zixiang Xu, Yanbo Wang, Yue Huang, Jiayi Ye, Haomin Zhuang, Zirui Song, Lang Gao, Chenxi Wang, **Zhaorun Chen**, Yujun Zhou, Sixian Li, Wang Pan, Yue Zhao, Jieyu Zhao, Xiangliang Zhang, and Xiuying Chen, [Modeling, Evaluating, and Enhancing Reasoning of Large Language Models in Complex Social Environments](#), in Proceeding of 2026 Conference on Empirical Methods in Natural Language Processing (**EMNLP 2026**), Budapest, Hungary, Oct 2026. [\[Red-Teaming Multi-Agent\]](#)
- [42] Zhihong Zhu, Kefan Shen, **Zhaorun Chen**, Yunyan Zhang, Yuyan Chen, Xiaoqi Jiao, Zhongwei Wan, Shaorong Xie, Wei Liu, Xian Wu, Yefeng Zheng, [DGLF: A Dual Graph-based Learning Framework for Multi-modal Sarcasm Detection](#), in Proceeding of 2024 Conference on Empirical Methods in Natural Language Processing (**EMNLP 2024**), Miami, Florida, Nov 2024. [\[Multimodal Safety\]](#)
- [43] Siyue Wang, **Zhaorun Chen**, Zhuokai Zhao, Chaoli Mao, Yiyang Zhou, Jiayu He, Albert Sibio Hu, [EscIRL: Evolving Self-Contrastive IRL for Trajectory Prediction in Autonomous Driving](#), in Proceeding of 8th Annual Conference on Robot Learning (**CoRL 2024**), Munich, Germany, Nov 2024. [\[Paper\]](#) [\[Code\]](#) [\[RL for Robotics\]](#)

ACADEMIC SERVICES

Conference Area Chair

- EMNLP 2025

Conference & Journal Reviewer

- NeurIPS, ICLR, COLM, ARR, IROS 2024
- ICLR, CVPR, ICML, NeurIPS 2025
- ICLR, CVPR, ICML, NeurIPS, Nature 2026

Organizer

- NeurIPS CLAS 2024: The Competition for LLM and Agent Safety [\[Link\]](#) 2024
- COLM 2025 Workshop on AI Agents: Capabilities and Safety (AIA 2025) [\[Link\]](#) 2025